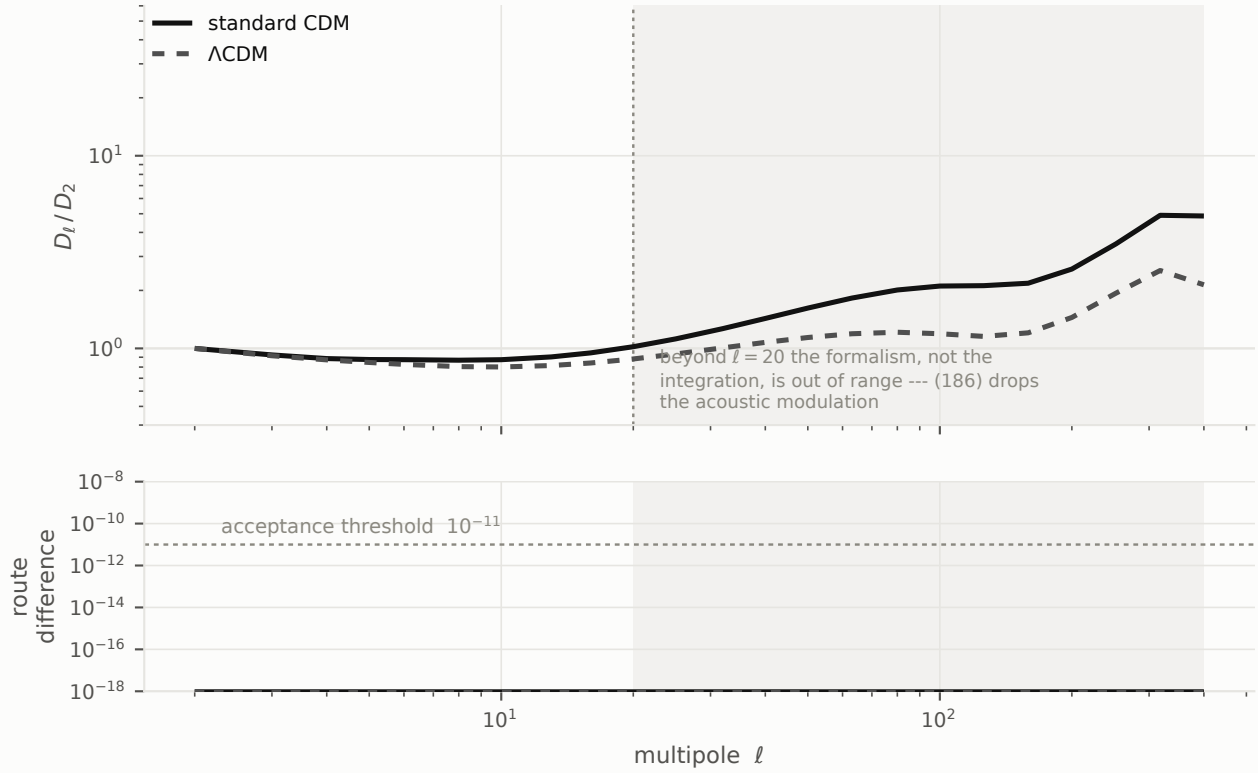
F3 The angular power spectrum from Annals II (186), by two routes

(186) computed from (176). The lower panel is the fractional difference between the mode route (186) and the covariant route (187)+(188), which are identical analytically — it is a check on the arithmetic, not on the physics.



CLAIMED OVER $2 \le \ell \le 20$ TO 5%, and shaded beyond. The binding limit is the formalism, not the integration: (186) drops the acoustic modulation, which costs 4.75% at $\ell = 20$ and grows. The k-grid truncation is looser at this tolerance, reaching 5% at $\ell \simeq 50$; at 1% the two invert. Recombination is equilibrium Saha. Criterion 5b, against CAMB and CLASS, is v1.2.0.